

Vinh Ly

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Summary

Software Engineer with **18 years** building distributed systems and large-scale infrastructure at Google, YouTube, and Whatnot. Architected systems processing billions of daily requests across search and video platforms serving **2B+ users**. Led engineering teams building real-time bidding for live commerce and real-time indexing for **search infrastructure**. Partnered with cross-functional teams on infrastructure strategy and technical direction. Virginia Tech honors graduate with **B.S. Computer Science**. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS, SQL

Distributed Systems & Infrastructure: distributed systems, large-scale systems, search infrastructure, real-time systems, video infrastructure, CDN, caching

Data Processing: MapReduce, Bigtable, distributed storage, data pipelines, streaming data

Programming: C++, Java, Python, Go

Infrastructure & Cloud: Kubernetes, GCP, CI/CD, Borg, on call

Experience

Education

B.S. Honors degrees in Computer Science

Virginia Tech | 2004 - 2007

Staff Software Engineer | YouTube

02/2025 - Present

- Architected **video infrastructure** serving billions of daily requests across storage and delivery pipelines processing **multi-petabyte** content through **CDN** for 2B+ users.
- Optimized **video streaming** systems reducing latency and improving playback quality for live and VOD content at massive scale.
- Led cross-functional initiatives on infrastructure improvements affecting platform reliability and performance across YouTube.
- Mentored engineers on distributed systems design, performance optimization, and career growth trajectories.

Staff Software Engineer | Whatnot

05/2024 - 02/2025

- Built **real-time bidding** infrastructure for live commerce platform supporting millions of daily transactions with **end-to-end** ownership.
- Established engineering processes and technical standards as team grew from 20 to 150 engineers including hiring and onboarding.
- Designed **auction engine** handling 500K+ concurrent users during peak live events with sub-second bid resolution.
- Created infrastructure reducing system costs by implementing efficient caching and resource optimization strategies.
- Defined technical roadmap and architecture for platform scalability supporting 10x traffic growth.

Staff Software Engineer/Manager | Google

03/2015 - 04/2024

- Led team of 8 engineers driving **technical strategy** for **search infrastructure** serving billions of users across product areas.
- Designed real-time indexing system processing millions of queries per second with 99.99% availability for core search serving.
- Developed **machine learning** models for **search ranking** improving relevance metrics across ranking systems.
- Mentored 8 engineers with 4 promoted to staff level through structured career development and technical guidance.

Software Engineer | Google

02/2011 - 02/2015

- Built **distributed data pipelines** processing billions of daily events supporting **search infrastructure** at scale.
- Developed large-scale data processing systems using **MapReduce** and **Bigtable** for search indexing workloads.
- Improved system reliability and reduced latency for search serving infrastructure used by billions of users.
- Contributed to search ranking infrastructure improving query processing efficiency and result quality.

Software Design Engineer II | Microsoft

10/2007 - 12/2011

- Designed and built **distributed storage** system handling significant data volume with end-to-end reliability requirements.
- Implemented software update distribution system serving millions of machines with 99.9% uptime.
- Improved system reliability metrics through **on-call** ownership and incident response for production systems.
- Collaborated cross-functionally on product launches and feature delivery for Windows platform services.